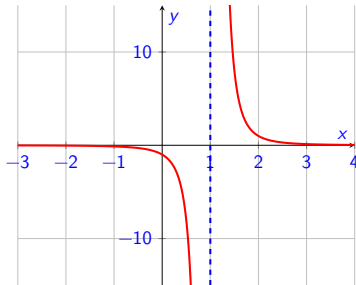


Week 03, Lecture 2 Vertical Asymptotes and Continuity

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This slides are by Niall Madden, with some content based on notes by Dr Kirsten Pfeiffer, and from Strang & Herman's "Calculus".

Reminders

- ▶ **Problem Set 1** due 5pm, Monday 28 September. You may access it multiple times, by clicking on [Assignments ... Problem Set 1 ...](#) and then, at the bottom of the page:
[Load Problem Set 1 in a new window](#)
- ▶ The **Tutorial Sheet** is available from Canvas.
- ▶ Problem Set 2 will open by Friday.
- ▶ The first (of two) class tests will take place Tuesday, 6th October.
If you wish to avail of Reasonable Accommodations for it, please complete this form:
<https://forms.cloud.microsoft/e/rBiq8RHHY2> by **10am, Tuesday 29 September**.

Today - a one-sided view of the following topics:

1 Infinite Limits

2 Digression: How fast can an object travel?

3 One-sided Limits

■ Notation

■ Piecewise functions

■ Empty and full circle notation

■ Existence of a limit

4 Vertical Asymptotes

■ Horizontal Asymptotes

5 Exercises

For more, see Chapter 2 (Limits) in **Calculus** by Strang & Herman. See openstax.org/books/calculus-volume-1/pages/2-introduction

In particular §2.2: One-sided limits (§2.2.4+§2.2.5) and vertical asymptotes (§2.2.7).

Infinite Limits

So far, we've had lots of examples that are a little like:

$$\lim_{x \rightarrow 1} \frac{x^3 - x^2 - x + 1}{(x - 1)^2} = 2.$$

(Check that this is correct).

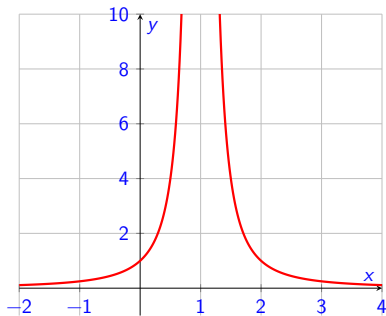
But what about

$$\lim_{x \rightarrow 1} \frac{1}{(x - 1)^2} = ???$$

Let's plot it and see:

Infinite Limits

$$f(x) = \frac{1}{(x-1)^2}$$



As x get closer and closer to 1, the value of $f(x)$ gets larger and larger. In fact, it becomes infinite.

For this we write

$$\lim_{x \rightarrow 1} f(x) = \infty.$$

Digression: How fast can an object travel?

- ▶ Q: Is there any limit to the speed at which an object can travel?
- ▶ A: Yes! (Assuming you believe Einstein)

Thanks to Einstein ($E = mc^2$), Lorenz and others, it is known that the mass of a moving charged particle behaves like

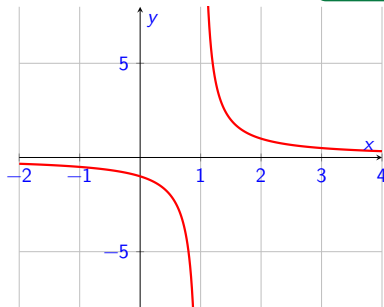
$$m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

where m_0 is its mass at rest, c is the speed of light, and v is the particles current speed. What happens as $v \rightarrow c$?

One-sided Limits

Let's consider a motivating example, very similar to the one where we introduced ∞ .

$$f(x) = \frac{1}{x-1}$$



As x get closer and closer to 1, then $f(x) \rightarrow -\infty$ or $f(x) \rightarrow \infty$, depending on whether x approaches 1 from the left or right.

To express this, we need the concept of a **one-sided limit**

$\lim_{x \rightarrow a^-} f(x)$ is: **limit of f as x approaches a from the left**

$\lim_{x \rightarrow a^+} f(x)$ is: **limit of f as x approaches a from the right**

These mean that

- ▶ if $\lim_{x \rightarrow a^-} f(x) = L$, then we can make $f(x)$ as close to L as we would like by taking $x < a$ as close to a as needed.
- ▶ If $\lim_{x \rightarrow a^+} f(x) = L$, then we can make $f(x)$ as close to L as we would like by taking $x > a$ as close to a as needed.

$\lim_{x \rightarrow a^-} f(x)$ is: **limit of f as x approaches a from the left**

$\lim_{x \rightarrow a^+} f(x)$ is: **limit of f as x approaches a from the right**

In the previous example, with $f(x) = \frac{1}{x-1}$, we have

► $\lim_{x \rightarrow 1^-} f(x) = -\infty$

► $\lim_{x \rightarrow 1^+} f(x) = \infty$

In many important examples, we encounter functions that have different definitions in different regions. The most classic example is the **absolute value function**:

$$|x| = \begin{cases} -x & x < 0 \\ x & x \geq 0. \end{cases}$$

Care has to be taken when evaluating the limits of such functions....

Example

Sketch the function

$$f(x) = \begin{cases} 3 - x, & x \leq 2 \\ \frac{x}{2} + 1, & x > 2 \end{cases}$$

Find $\lim_{x \rightarrow 2^-} f(x)$ and $\lim_{x \rightarrow 2^+} f(x)$.

Empty and Full Circle Notation:

In the previous sketch, we use the convention that

- ▶ If the end point of a line segment is **not** included in its definition, it terminates with an **open circle**, $\circ$
- ▶ If the end point of a line segment **is** included in its definition, it terminates with an **closed circle** $\bullet$.

Existence of a limit

$\lim_{x \rightarrow a} f(x)$ **exists** if and only if

$$\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x)$$

So if $\lim_{x \rightarrow a} f(x) = L$ exists, we have

$$\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a^+} f(x) = L$$

though it is not necessary that $f(a) = L$

Example

Sketch the function

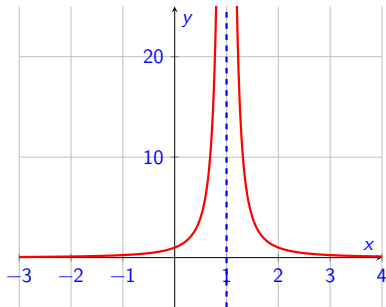
$$f(x) = \begin{cases} 3 - x, & x < 2 \\ 4, & x = 2 \\ \frac{x}{2}, & x > 2 \end{cases}$$

Determine if $\lim_{x \rightarrow 2} f(x)$ exists.

Vertical Asymptotes

Let's revisit an example from earlier:

$$f(x) = \frac{1}{(x-1)^2}$$



Note that the points on the graph having x -coordinates very near to 1 are very close to the vertical line $x = 1$. That is, as x approaches 1, the points on the graph of $f(x)$ are closer to the line $x = 1$.

We call the line $x = 1$ a **vertical asymptote** of the graph.

Vertical Asymptotes

Definition: Vertical Asymptote

The vertical line $x = a$ is a **vertical asymptote** of $f(x)$ if any of $\lim_{x \rightarrow a^-} f(x)$, $\lim_{x \rightarrow a^+} f(x)$, or $\lim_{x \rightarrow a} f(x)$ are ∞ or $-\infty$.

To find a vertical asymptote of a function $f(x) = \frac{p(x)}{q(x)}$, we find a value, a for which $p(a) \neq 0$ but $q(a) = 0$.

Vertical Asymptotes

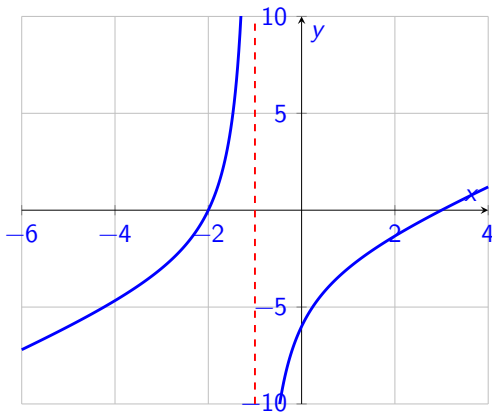
Example

Find any vertical asymptotes of

$$f(x) = \frac{x^2 - x - 6}{x + 1}$$

Vertical Asymptotes

$$f(x) = \frac{x^2 - x - 6}{x + 1}$$



Vertical Asymptotes

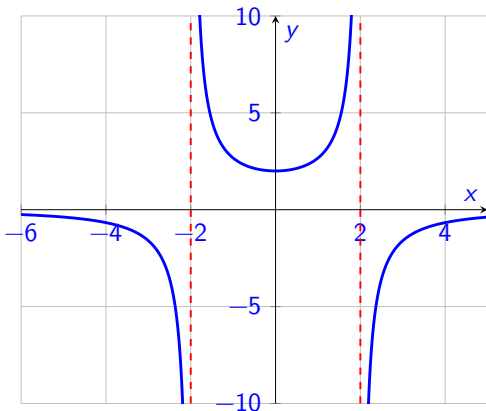
Example

Find all vertical asymptotes of the graph of

$$g(x) = -\frac{8}{x^2 - 4}.$$

Vertical Asymptotes

$$f(x) = -\frac{8}{x^2 - 4}$$



There is a related concept of a **horizontal asymptote**, but we'll save that for later, when we cover “limits at infinity”.

Exercise 3.2.1

Find all the vertical asymptotes of $f(x) = \frac{x+2}{x^2+2x-8}$.

For more practice try the Exercises at the end of

<https://openstax.org/books/calculus-volume-1/pages/2-3-the-limit-laws>, especially Problems 93–103.